FLEET TEAM DIGEST

737MAX-FTD-26-17001

Issue Title : Halon Replacement

Airplane Model	ATA	Minor Model(s)
737MAX	2600-00	-8, -8200, -9, -BBJ

Other Model(s)	ECCN
DC-2/-3/-4/-6/-7, MD-80, 767, 737, DC-10, DC-9, 777, 777X, 727, 717, 737NG, 747, 787, MD-11, 757, 707, DC-8, MD-90	9E991

Originated Date	Last Revised Date	Created On
05/26/2017	08/28/2024	08/14/2024

Estimated Completion Date	Next Update Date
12/31/2025	01/17/2025

Revision Description

No changes for this revision. The next significant milestone is expected at the end of 2024, which is for the Halon Alternatives for Aircraft Propulsion Systems (HAAPS) Phase III final agent selection.

Applicability

All Boeing airplanes using Halon 1211 and 1301 as a fire extinguishing agent.

Description

This FLEET TEAM Digest (FTD) provides information regarding International Civil Aviation Organization (ICAO) requirements and governmental regulations for Halon fire extinguishing agent replacement.

Halon has been used as an effective fire extinguishing agent on aircraft since the 1960s in lavatories, cabins, and flight decks (handheld fire extinguishers), engines and auxiliary power units (APUs), and cargo compartments. The production of Halon, an ozone-depleting substance, has been phased out since 1994 under the requirements of the United Nations Montreal Protocol on Substances that Deplete the Ozone Layer. However, due to the safety-critical nature of aviation, a Halon “critical use” exemption has been in place.

International Civil Aviation Organization (ICAO) has established deadlines for implementing Halon replacements.

The International Coordinating Council of Aerospace Industries Associations (ICCAIA) have met to discuss the resolution of Halon replacement. The industry has been active in researching Halon alternatives and in working with suppliers and regulatory agencies to address all associated safety, environmental, and operational requirements.

Background

General - In September 2013, the European Union (EU) EASA initiated a rulemaking group (RMT.0560) with the objective to amend

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Part-26 and CS-26 (originally proposed by NPA 2012-13) in order to comply with Amendments to Annex 6 for lavatory waste receptacles and handheld fire extinguishers in cabins. The Notice of Proposed Amendment (NPA) has completed the Final Comment-Response Document (CRD) phase, and an EASA Opinion has been submitted recommending an amendment to Annex I (Part-26) to Regulation (EU) 2015/640 — "Additional airworthiness specifications".

Reference URL links:

http://easa.europa.eu/document-library/notices-of-proposed-amendment/npa-2014-26

http://www.easa.europa.eu/system/files/dfu/CRD%202014-26.pdf (CRD)

http://www.easa.europa.eu/document-library/opinions/opinion-082016 (EASA Opinion)

In November 2013, the US FAA initiated a Halon Replacement Aviation Rulemaking Committee to make recommendations on the transition from Halon to alternative agents by reviewing the safety, environmental, and economic issues and taking into account the ICAO timelines.

Reference URL link: http://www.faa.gov/regulations_policies/rulemaking/committees/documents/index.cfm/committee/browse/committeeID/397#tabs-1.TR

Cargo Systems - In 2013, ICCAIA formed a Cargo Compartment Halon Replacement Working Group (CCHRWG) to provide a recommendation to ICAO regarding a cargo compartment Halon replacement deadline and implementation period for new type certification aircraft applications. At the 39th Assembly, ICAO accepted industry's recommendation for a 2024 cargo compartment deadline (new TC application); this was recently adopted into Annex 8 (April 2018). (ref http://www.icao.int/Meetings/a39/Documents/WP/wp_514_en.pdf).

Propulsion Systems - The industry has formed the Halon Alternatives for Aircraft Propulsion Systems (HAAPS) consortium on engine and APU Halon replacements. Reference URL link for progress updates:

https://www.fire.tc.faa.gov/pdf/systems/Aug20Meeting/Macias-0820-HAAPS.pdf

Hand-held Extinguishers – ICAO Annex 6 (Assembly Resolution A37-9) had specified 31 December 2016 as the deadline for delivery of new aircraft containing Halon 1211 extinguishers; this standard was adopted by a number of countries. Through the ICCAIA, the ICAO was petitioned at the 39th Assembly to extend this deadline to 31 December 2018; this revision was adopted via Amendment 41 to Part 1 of Annex 6, and went into effect on 27 April 2017. US EPA SNAP approval for the alternative agent 2-BTP was granted on 26 September 2016.

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Lavatory Fire Extinguishers - A retrofit solution is not required for 737MAX lavatories since all 737MAX aircraft have been delivered with the non-Halon extinguisher since the beginning of the program.

EASA and the EC have published a guide to explain both the EASA implementation of ICAO Standards regarding Halon and the EC implementation of the Montreal Protocol. This guide for EASA/EC Halon Replacement Implementation can be found in the attachments section. Table 3 in the guide provides a detailed summary of deadlines for Halon Replacement.

Status

Hand-Held Fire Extinguishers - The 2-BTP extinguishers (designated as “Halotron BrX”) are now available for customer selection. Customer Engineering Airplane Configuration Bulletin 2019-ACB-0022 was published on 29 July 2019 for airplane model applicability for non-Halon extinguishers. Please see the airplane catalog for the latest options for non-Halon extinguishers. In-service replacement extinguishers will be available through Boeing Materials Management following initial delivery. EASA has officially mandated the ICAO rule that all new deliveries under EASA are required to have non-Halon portable extinguishers starting from 18 May 2019. This mandate also applies to third-country operators who operate into the EU.

The ICAO standards are a “forward fit” requirement, applying only to aircraft manufactured after a certain date. For example, EASA implemented their rule for hand-held extinguishers only for aircraft manufactured on or after 18 May 2019. However, the EC Ozone regulation will require retrofit of the hand-held extinguishers on all airplanes operating in Europe at the end of 2025. This is clarified in the EASA/EC Halon Replacement Implementation Guide.

The Civil Aviation Administration of China (CAAC) issued an airworthiness directive (AD) on 25 November 2021. The AD is effective immediately, and mandates that Halon portable fire extinguishers cannot be used in all airplanes applying for CAAC registration and a CAAC airworthiness certificate starting from 25 November 2021. Airplanes that applied for CAAC registration and a CAAC airworthiness certificate between 31 December 2018 and 25 November 2021 must replace any Halon portable fire extinguishers with non-Halon portable fire extinguishers by 31 May 2022.

Engine/APU and Cargo Compartment - While no Halon replacement solutions are currently available for Engine/APU or Cargo Compartment, Boeing continues to research and monitor progress on alternative agents/approaches, supply chain readiness, testing, qualification, and certification to support the EC and ICAO deadlines. The EU Regulation 2017/605 was released March 2017, clarifying the definition of ‘new equipment’ from a type certification perspective. Amended Type Cert applications are thus excluded from EU 1005/2009 deadlines for Halon replacement.

HAAPS has completed a detailed review of alternative proposals submitted by various prospective suppliers (firex system suppliers, firex agent suppliers, firex distributors, chemical companies). The outcome of those reviews was down selection to two primary candidate alternatives for evaluation with additional contingency alternatives for possible evaluation in the final phase of HAAPS’ activities, Phase III, which formally commenced in October 2020.

Work in-progress to support Phase III initiation includes development of supplier engagement for non-technical documentation, technical validation planning, defining applicable regulatory requirements and certification plans, and next engagement with airworthiness authorities. Final agent down selection(s) and completion of HAAPS activities are expected by the end of 2024.

Interim Action

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Please see the airplane catalog for the latest options for non-Halon hand-held fire extinguishers.

Final Action

Boeing to provide Halon replacement solutions for the Portable Hand-Held Fire Extinguishers, Engine and APU, and Cargo Compartment fire extinguisher systems.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

- Production Incorporation of Alternative Lavatory Agent: 2006 Start-point (complete)

- Industry Cargo Compartment Halon Replacement Working Group Established: March 2013 (complete)

- EU EASA “Halon Replacement” Rule-making Group (RMT.0560) Established: September 2013 (complete)

- ICAO reaffirms Halon replacement resolution: October 2013 (complete)

- US FAA “Halon Replacement” Rule-making Committee Established: November 2013 (complete)

- Production Incorporation of Alternative Hand-Held Agent: December 2017 Start-point (complete)

- HAAPS Joint Collaboration Agreement Signed by Founding Members: June 2018 (complete)

- HAAPS Consortium Formed: October 2018 (complete)

- ACB 2019-ACB-0022 Released: 29-July-2019 (complete)

- HAAPS Phase II Complete: 30-September-2020 (complete)

- HAAPS Phase III Complete: December 2024 (target)

- EASA Portable Fire Extinguisher Halon Replacement: December 2025 (target)

Reference the Illustrations section where the Halon replacement solution(s) deadlines are identified.

Operator Action

Operators who are required to comply with the CAAC AD for the portable fire extinguisher retrofit by 31 May 2022 should contact the Boeing Global Services and Modifications Group to obtain a quote using the attached Request for Proposal Form.

The Halon Recycling Corporation (HRC) has published the attached reference guidance document for servicing of Halon 1301

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bottles. HRC includes airframe manufacturers, other users of Halon 1301, fire protection systems suppliers, and halon recycling companies. Halon 1301 bottles are removed for servicing every 5-12 years, and the Halon 1301 agent is removed from the bottle during servicing and replaced with recycled Halon 1301. Ideally, this process should recover a great majority of the Halon 1301. However, an FAA Aviation Rulemaking Committee estimated in 2008 that the amount lost each year could be as much as 7-8% of the total installed amount. The guidance document recently developed by HRC explains some of the possible explanations for this high rate, and best practices that can be instituted to increase the amount that is recovered. This will have multiple benefits, including minimizing environmental damage from Halon 1301 that is vented, lowering the cost of servicing, and extending the length of time that recycled Halon 1301 will be available for maintenance of the aviation fleet.

Boeing encourages airline maintenance engineering teams to review the HRC document and to share this information with MRO partners. The HRC website link is (https://www.halon.org/hrc-av-outreach).

Boeing would also like to hear from airlines about challenges experienced during the servicing of systems, and any data that can be shared about the amount of Halon 1301 that is recovered during servicing. Operators are requested to share this information with Boeing by submitting a Service Request via the Boeing Communication System (BCS).

Part Information

Lavatory

Kidde P/N G800100-3

Kidde P/N A800100-3

Pacific Scientific HTL P/Ns 30100022-3 and -33

Hand-Held

Kidde P/N 476288

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Other	Halon Recycling Corporation Guidance		
Other	ICAO Working Paper A37-WP/197		09/20/2010
Service Letter (SL)	777-SL-26-014	B	11/16/2017
Boeing AERO Magazine	Issue 44_Quarter 04/2011		
Other	Civil Aviation Department AN 60A Issue 1		03/30/2012
Other	CAAS AC SAR-1(1)		12/15/2016
Other	ICAO Working Paper A38-WP/36		06/06/2013
Other	Intl Coord Council Aero Ind Ass Wk Pap A38-WP/238		09/06/2013
Other	ICAO Working Paper AN-WP/9124		10/13/2016

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Other	ICAO State Letter AN 11/1.3/30-16/97		11/30/2016
Airplane Configuration Bulletin (ACB)	2019-ACB-0022		07/29/2019
Other	CAAC AD CAD2021-MULT-80		11/25/2021

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Hand Held Fire Bottle (2-BTP)		Kidde	476288
Lavatory Firex Bottle (FM200 Non-Halon Agent)		Kidde	G800100-3
Lavatory Firex Bottle (Halon 1301 Agent)		Kidde	A800100-3
Potty Bottle		Pacific Scientific HTL	30100022-3
Lavatory Fire Extinguisher		Pacific Scientific HTL	30100022-33

Related Categories

- Technical Focus Item : null

Attachments:

a) Request For Proposal Form

b) HRC Guidance for Reducing Halon Emissions

c) EASA/EC Halon Replacement Implementation Guide

d) Halon Timelines

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